Intro to HPC

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March 2025

This material is extended and adapted from

"How to start with spatial omics" module "Data Analysis" by Janick Mathys

HPC

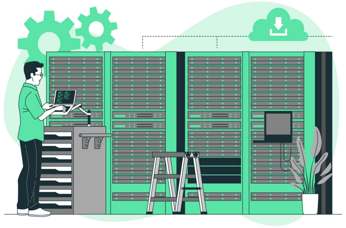
High-performance computing

High-performance computing

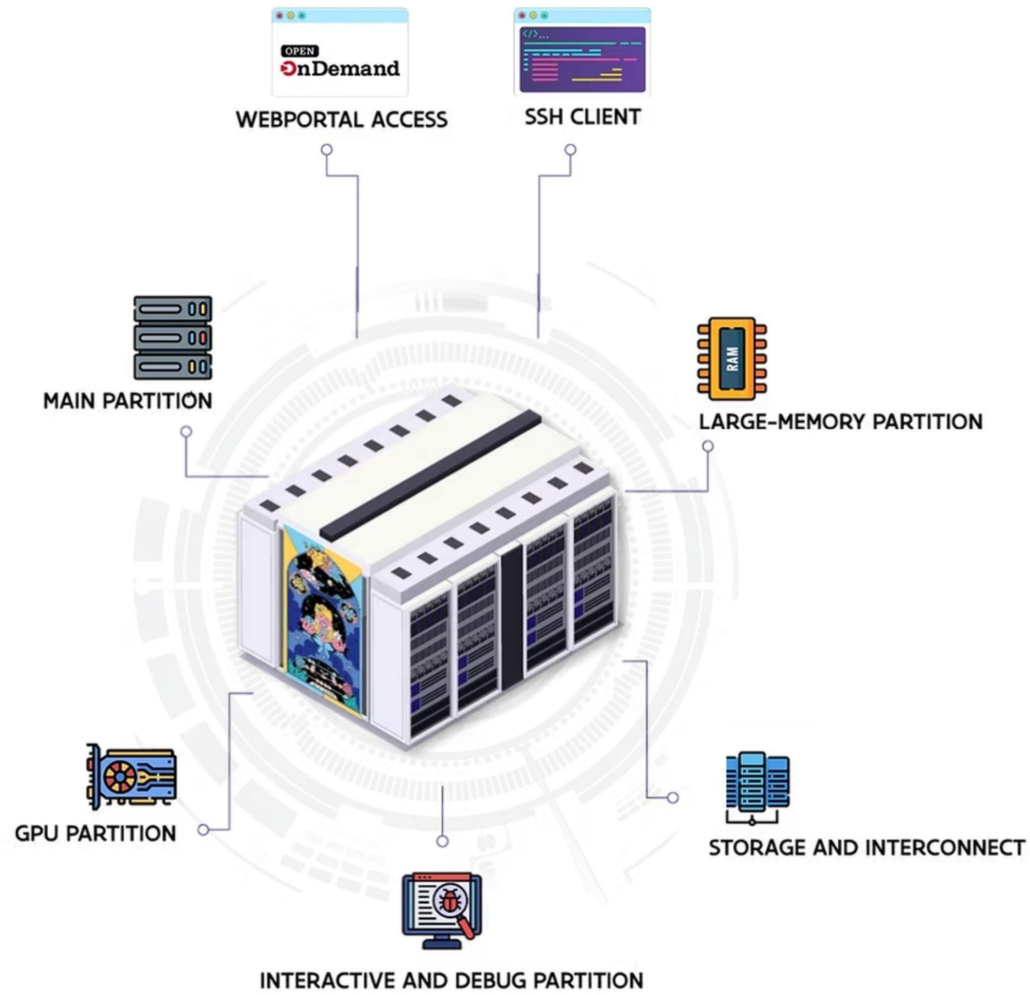
Computer

Server

HPC



- Different resources available
  - Nodes → parallel file system
    - CPU with
      - Multi-processor interconnect
      - High memory
    - GPU
      - Specialized processor
    - Fast interconnection

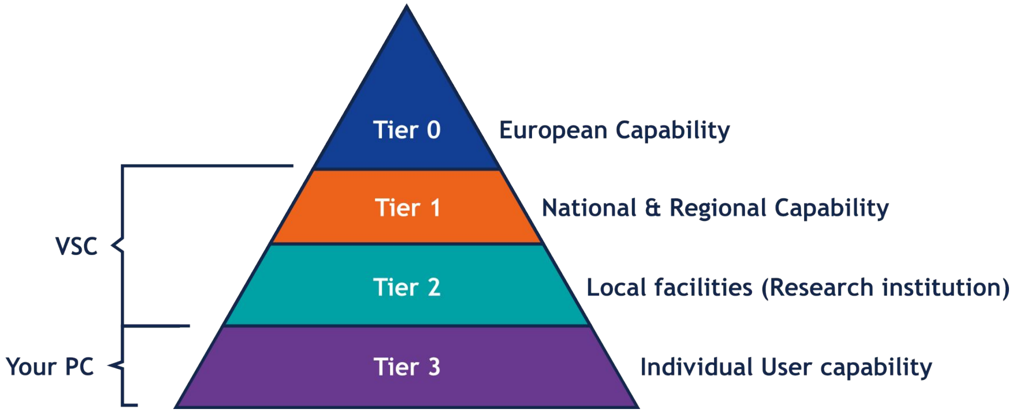

OPEN OnDemand
WEBPORTAL ACCESS
SSH CLIENT
MAIN PARTITION
RAM
LARGE-MEMORY PARTITION
GPU PARTITION
STORAGE AND INTERCONNECT
INTERACTIVE AND DEBUG PARTITION

HPC

Tiers & Resources

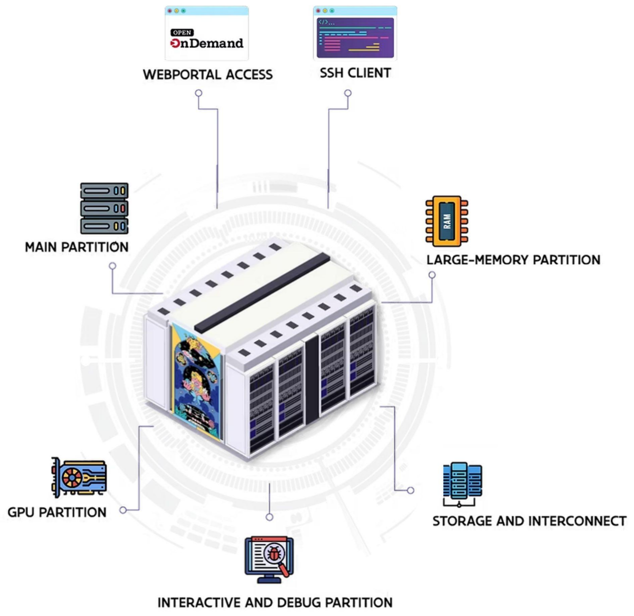
EU model: Infrastructure

- Tier-0
  - 6 in EU
  - Collaboration
- Tier-1
  - Hortense (UGent)
- Tier-2
  - Stevin (UGent)
  - Genius (KULeuven)
  - wICE (KULeuven)



VIB Data Core Compute Cluster

- A centralized Compute solution for VIB users
  - Similar infrastructure
  - Only for VIB employees
  - Not using the Tier system classification



HPC

File System

File System

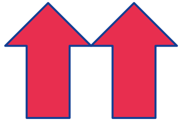
Purpose & Maintenance

Capacity (GB)

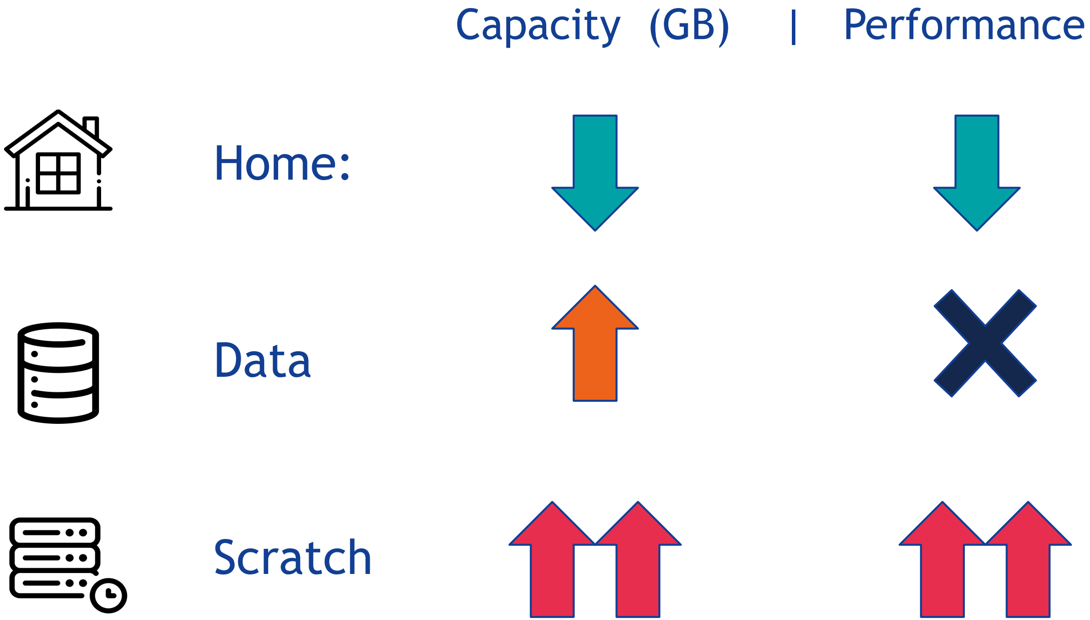
Home:

Data

Scratch



File System
Purpose & Maintenance



VIB

File System

Purpose & Maintenance

Capacity (GB) | Performance | Purpose

Home:

- Config and user preferences
- No job execution
- No data

Data

Scratch

File System
Purpose & Maintenance

	Capacity (GB)	Performance	Purpose
Home:			Config and user preferences No job execution No data
Data			Long-term storage No job execution
Scratch			

File System
Purpose & Maintenance

	Capacity (GB)	Performance	Purpose
Home:			▪ Config and user preferences ▪ No job execution ▪ No data
Data			▪ Long-term storage ▪ No job execution
Scratch			▪ Short-term storage ▪ Job executed ▪ Not accessible in other VSCs

File System
Purpose & Maintenance

	Capacity (GB)	Performance	Purpose
Home:	↓	↓	▪ Config and user preferences ▪ No job execution ▪ No data
Data	↑	✕	▪ Long-term storage ▪ No job execution
Scratch	↑↑	↑↑	▪ Short-term storage ▪ Job executed ▪ Not accessible in other VSCs

Copy
before run

File System
Different systems

$VSC_HOME

$VSC_DATA

$VSC_SCRATCH

Other Scratch

* $HOME = $VCS_SCRATCH

** Project based

File System
Different systems

	** Tier1 UNIVERSITEIT GENT	Tier2 UNIVERSITEIT GENT	** KU LEUVEN Tier 2
$VSC_HOME	* $HOME	3 GB	3 GB
$VSC_DATA	Variable	25 GB	75 GB
$VSC_SCRATCH	3 GB	25 GB	500 GB
Other Scratch	Variable	X	up to 600 GB
	$VSC_SCRATCH_PROJECTS_BASE		$VSC_SCRATCH_NODE

VIB

File System
Different systems

** Project based

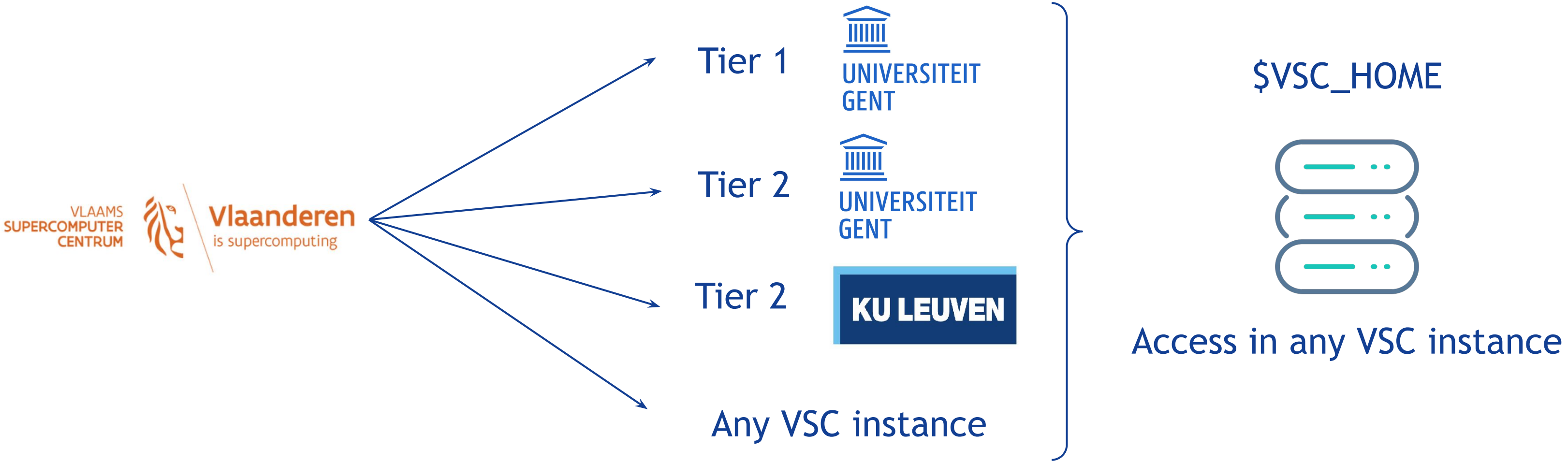
	** Tier1 UNIVERSITEIT GENT	Tier2 UNIVERSITEIT GENT	** KU LEUVEN Tier 2	** VIB Data Core compute	
$VSC_HOME	* $HOME	3 GB	3 GB	$HOME	10 GB
$VSC_DATA	Variable	25 GB	75 GB	path/to/	Variable
$VSC_SCRATCH	3 GB	25 GB	500 GB	X	X
Other Scratch	Variable	X	up to 600 GB	$SCRATCHDIR	Variable

$VSC_SCRATCH_PROJECTS_BASE

$VSC_SCRATCH_NODE

Local to Node

Accessible from:



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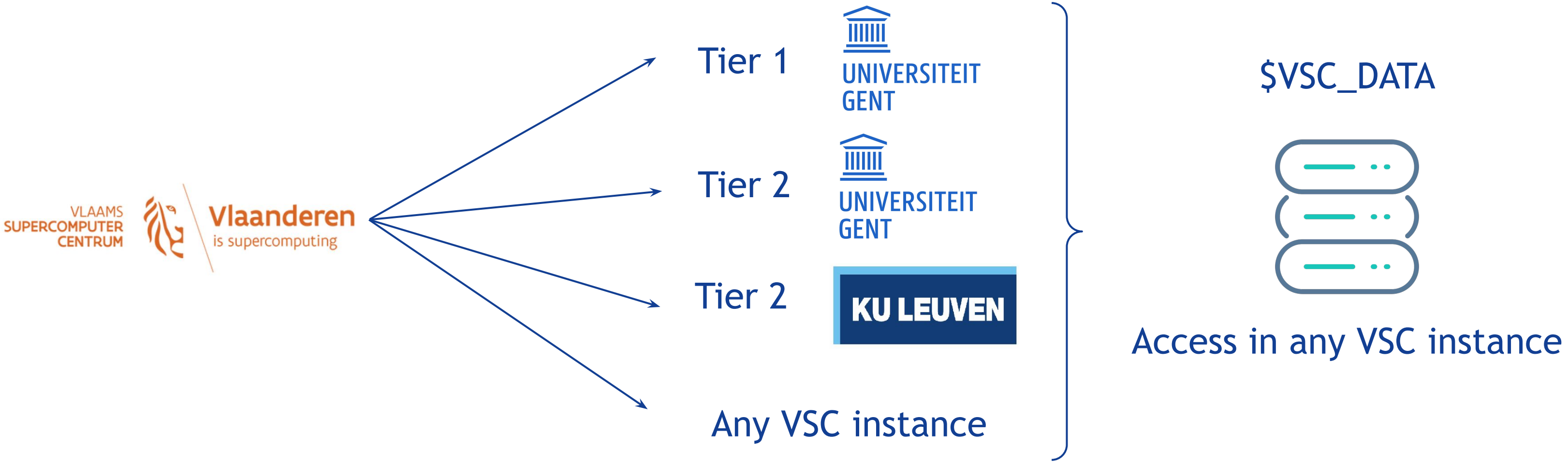

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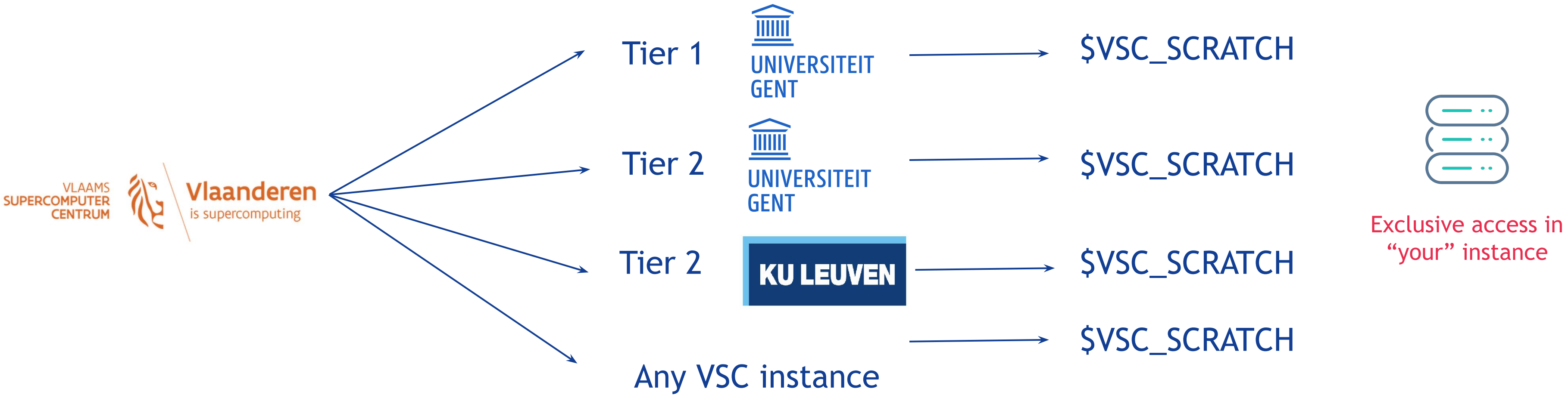


VIB

Accessible from:

VLAAMS SUPERCOMPUTER CENTRUM
Vlaanderen is supercomputing

Tier 1 — UNIVERSITEIT GENT

$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE

Tier 2 — UNIVERSITEIT GENT

$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE

Tier 2 — KU LEUVEN

$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE

Any VSC instance

$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE
$VSC_SCRATCH_NODE

Exclusive of each node

Similar for VIB Data Core compute

HPC

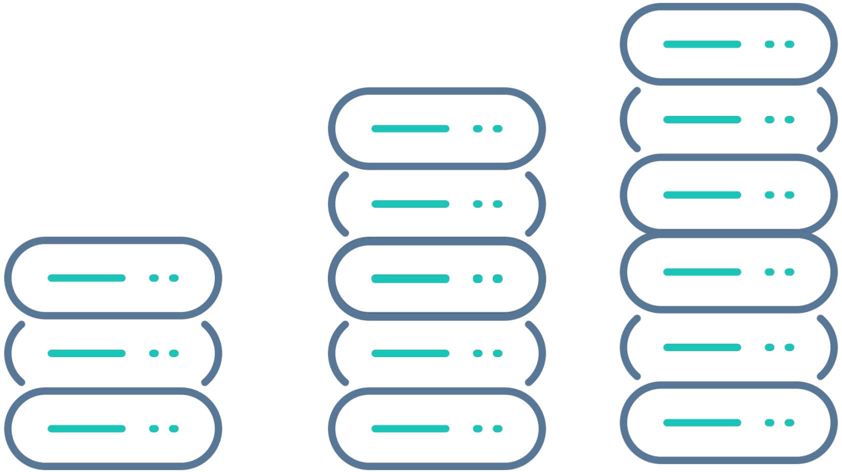
Node offer

Nodes in HPC

- Variable resources
  - CPUs
  - GPUs
  - Memory

- Debug interactive node

- Each node has a name



Nodes in HPC

- Variable resources
  - CPUs
  - GPUs
  - Memory

More resources == Less priority

- Debug interactive node

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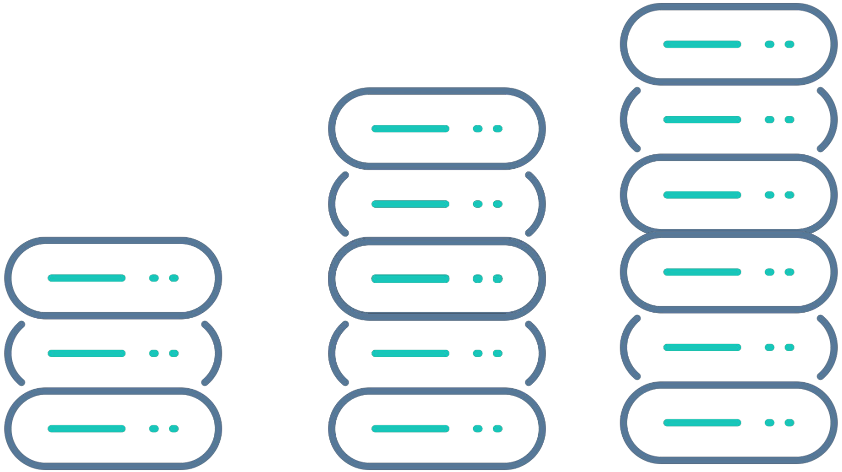

Nodes in HPC

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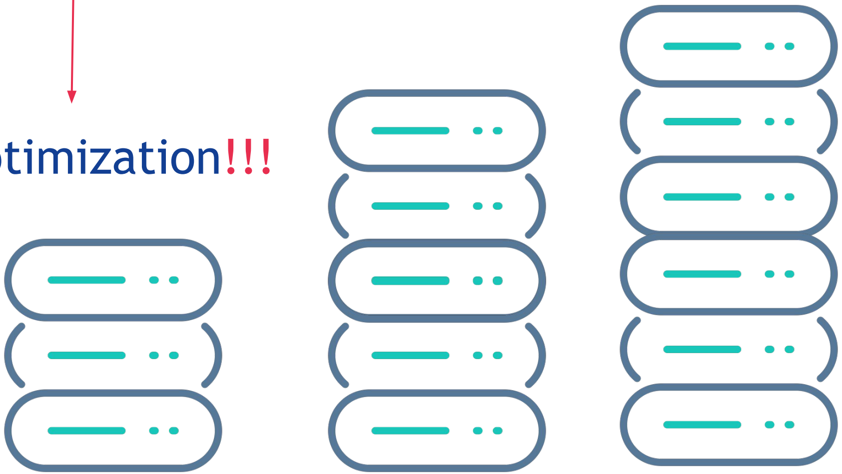
- Debug interactive node

Optimization!!!

- Each node has a name



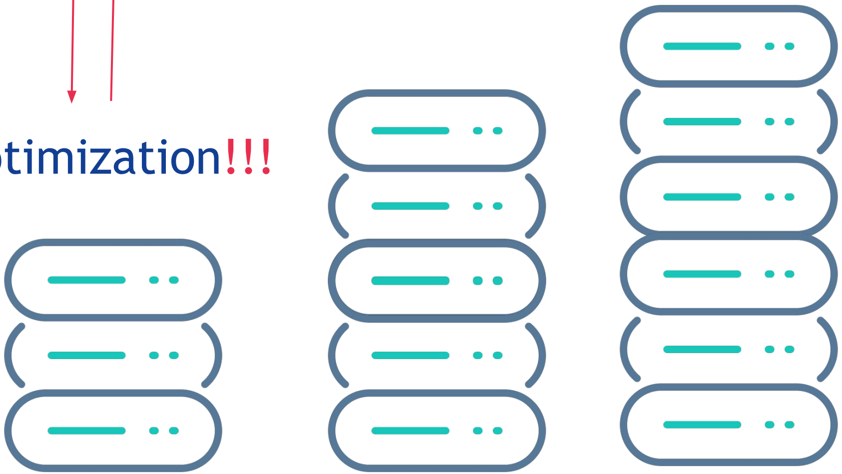
Nodes in HPC

- Variable resources
  - CPUs
  - GPUs
  - Memory

Look into *log file*

jobname.logID

Optimization!!!



Nodes and names

debug_28C_56T_750GB

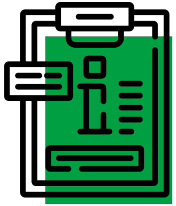
[VIB]

donphan

[Tier 2 UGent]

debug_rome

[Tier 1 UGent]



interactive
gpu_a100_debug

[Tier 2 KULeuven]

Nodes and names

debug_28C_56T_750GB

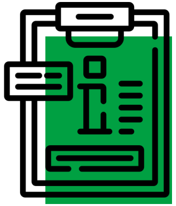
[VIB]

donphan

[Tier 2 UGent]

debug_rome

[Tier 1 UGent]



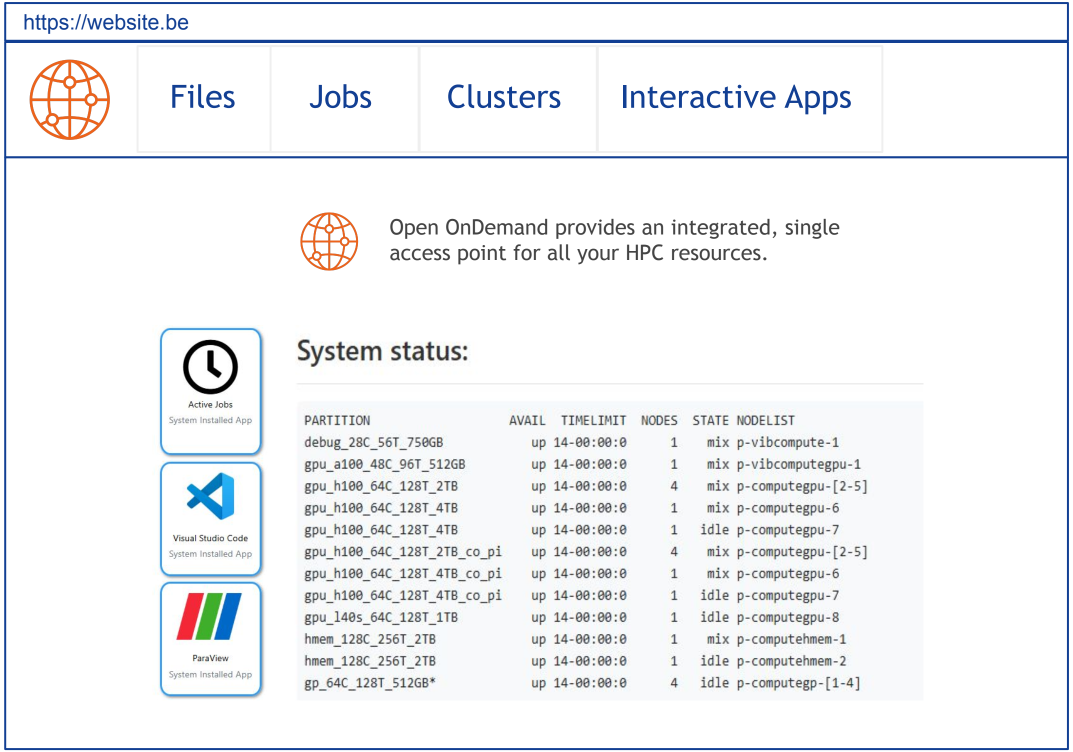
interactive
gpu_a100_debug

[Tier 2 KULeuven]

HPC

Connect to OnDemand

OnDemand: Friendly interface



PARTITION	AVAIL	TIMELIMIT	NODES	STATE	NODELIST
debug_28C_56T_750GB	up	14-00:00:0	1	mix	p-vibcompute-1
gpu_a100_48C_96T_512GB	up	14-00:00:0	1	mix	p-vibcomputegpu-1
gpu_h100_64C_128T_2TB	up	14-00:00:0	4	mix	p-computegpu-[2-5]
gpu_h100_64C_128T_4TB	up	14-00:00:0	1	mix	p-computegpu-6
gpu_h100_64C_128T_4TB	up	14-00:00:0	1	idle	p-computegpu-7
gpu_h100_64C_128T_2TB_co_pi	up	14-00:00:0	4	mix	p-computegpu-[2-5]
gpu_h100_64C_128T_4TB_co_pi	up	14-00:00:0	1	mix	p-computegpu-6
gpu_h100_64C_128T_4TB_co_pi	up	14-00:00:0	1	idle	p-computegpu-7
gpu_l40s_64C_128T_1TB	up	14-00:00:0	1	idle	p-computegpu-8
hmem_128C_256T_2TB	up	14-00:00:0	1	mix	p-computehmem-1
hmem_128C_256T_2TB	up	14-00:00:0	1	idle	p-computehmem-2
gp_64C_128T_512GB*	up	14-00:00:0	4	idle	p-computegp-[1-4]

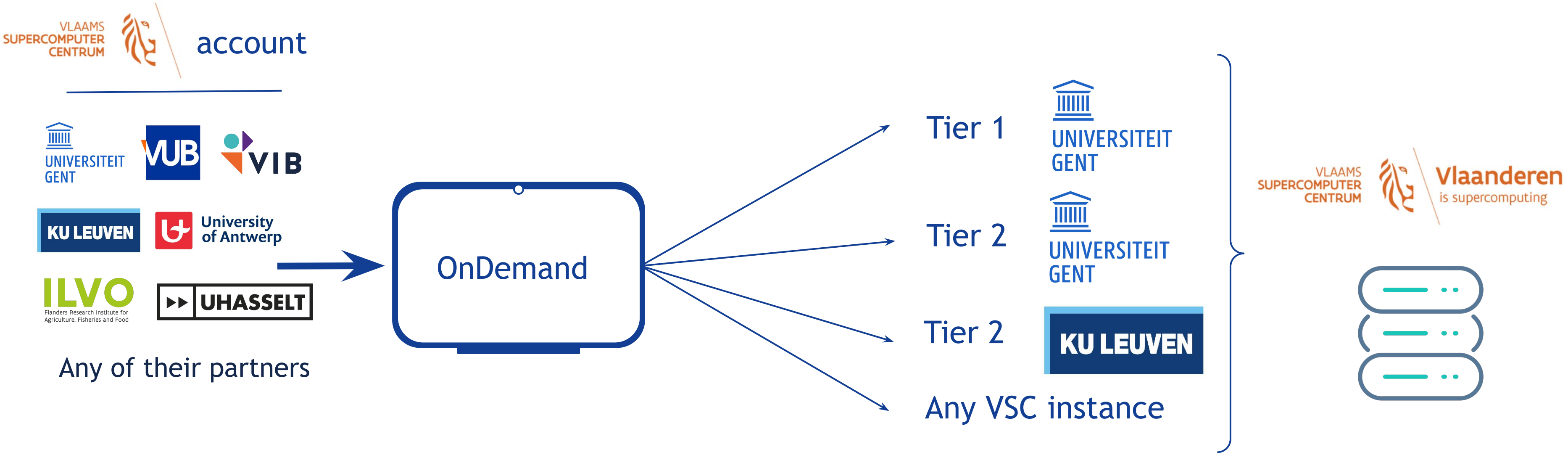
VIB
https://compute.vib.be

UGent Tier-2
https://login.hpc.ugent.be

UGent Tier-1
https://tier1.hpc.ugent.be/

KULeuven Tier-2
https://ondemand.hpc.kuleuven.be/

Where can you connect ?



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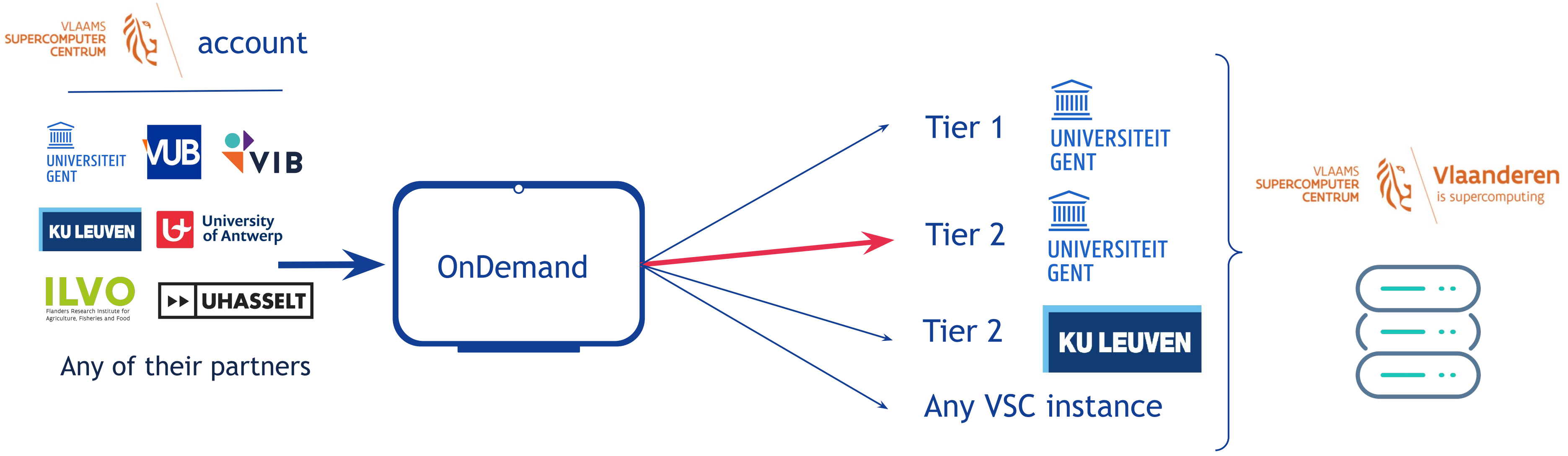

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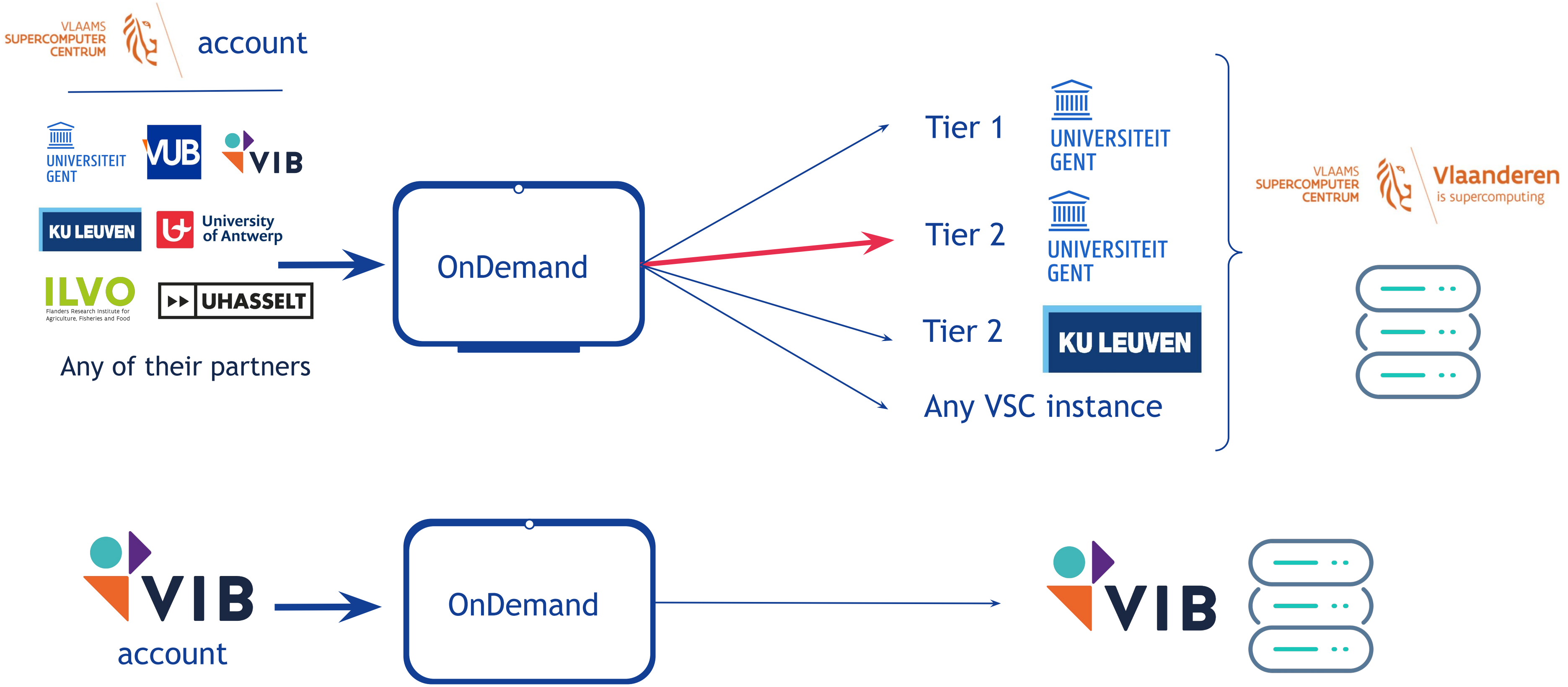
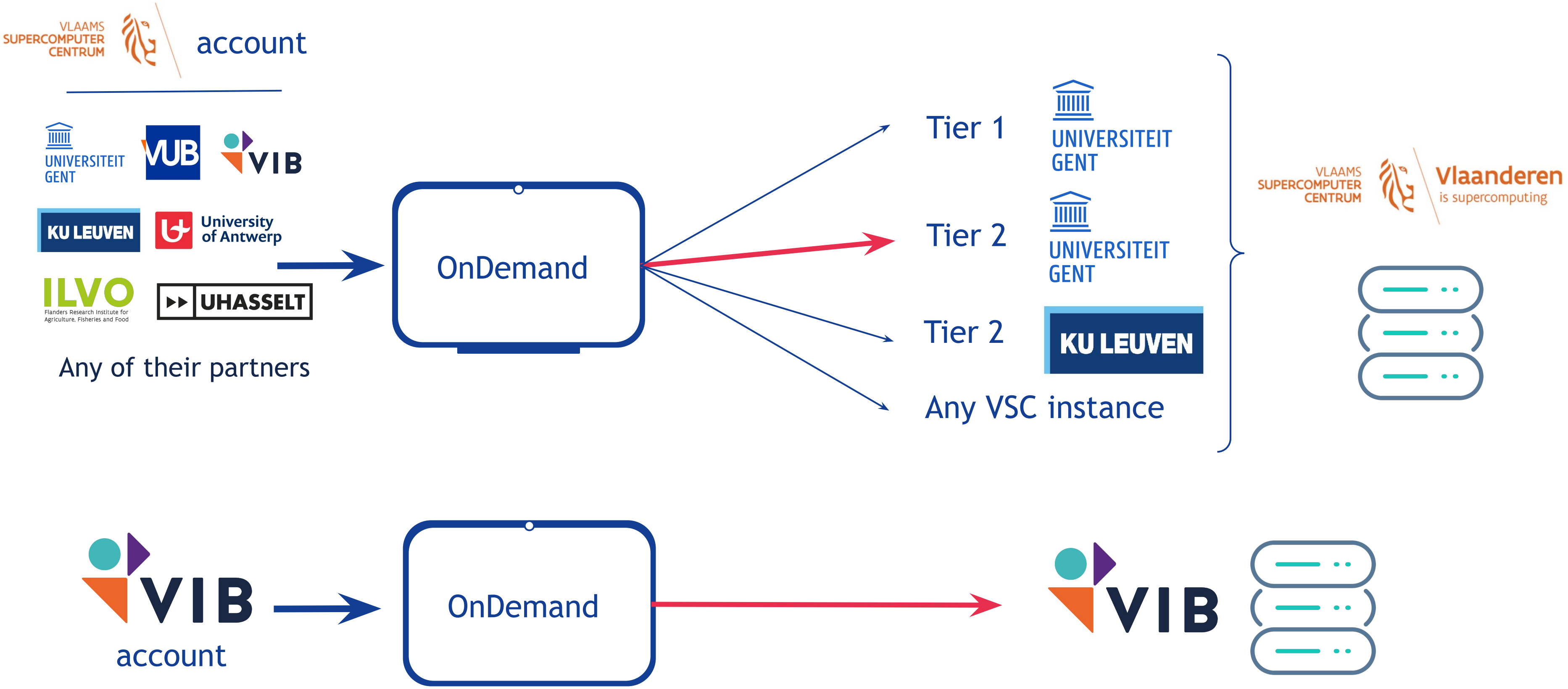


Where can you connect ?



Let’s connect

Chose one or both:

- VIB
- UGent Tier-2

Steps to follow:

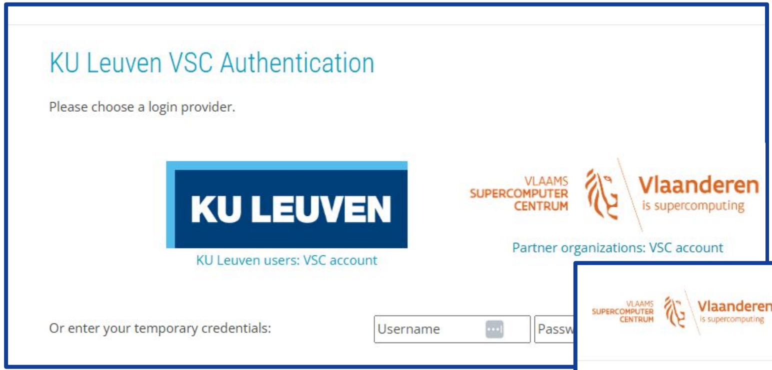
Let’s connect

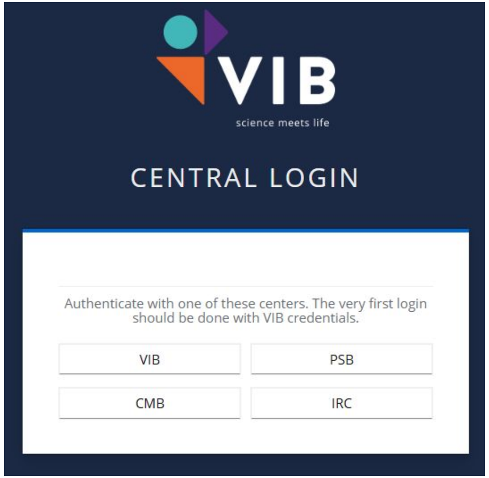
Chose one or both:

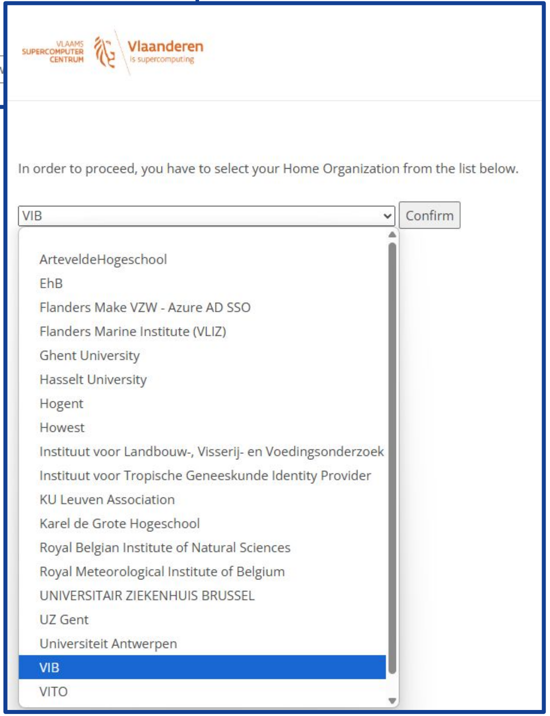
[VIB]
[UGent Tier-2]

Steps to follow:

- Choose your institutions







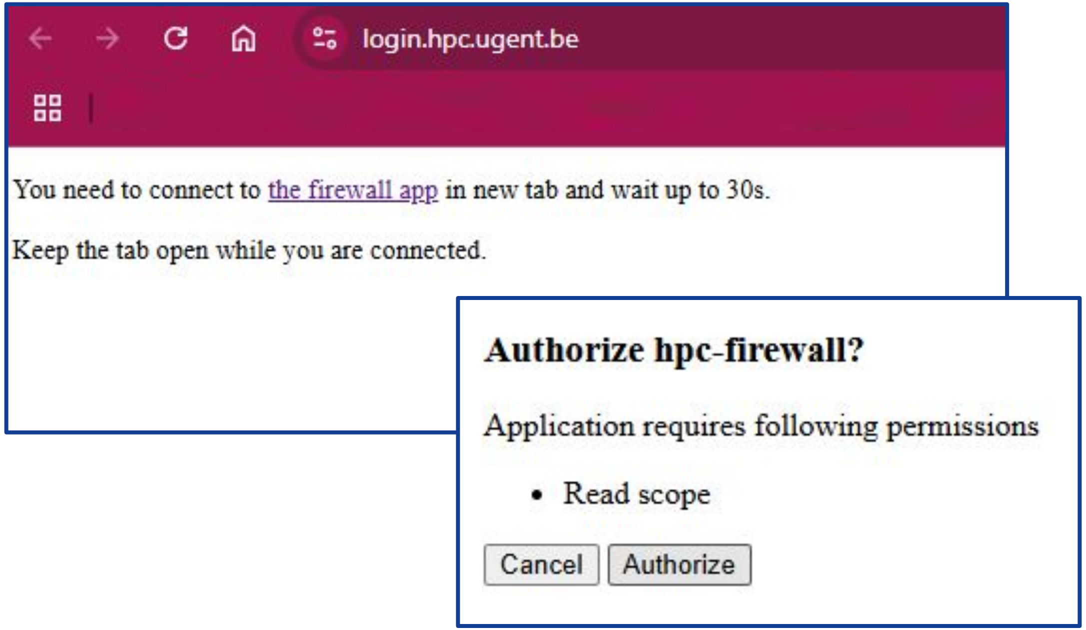
Let’s connect

Chose one or both:

[VIB]()
[UGent Tier-2]()

Steps to follow:

- Choose your institutions
- Authorize access



HPC

Connect cmd-line

Create an SSH key

- Check existing keys

```
ls ~/.ssh
```

- Generate a new key just for VSC

```
ssh-keygen -t ed25519 -C "your_email@example.com" -f ~/.ssh/ed25519_vsc
```

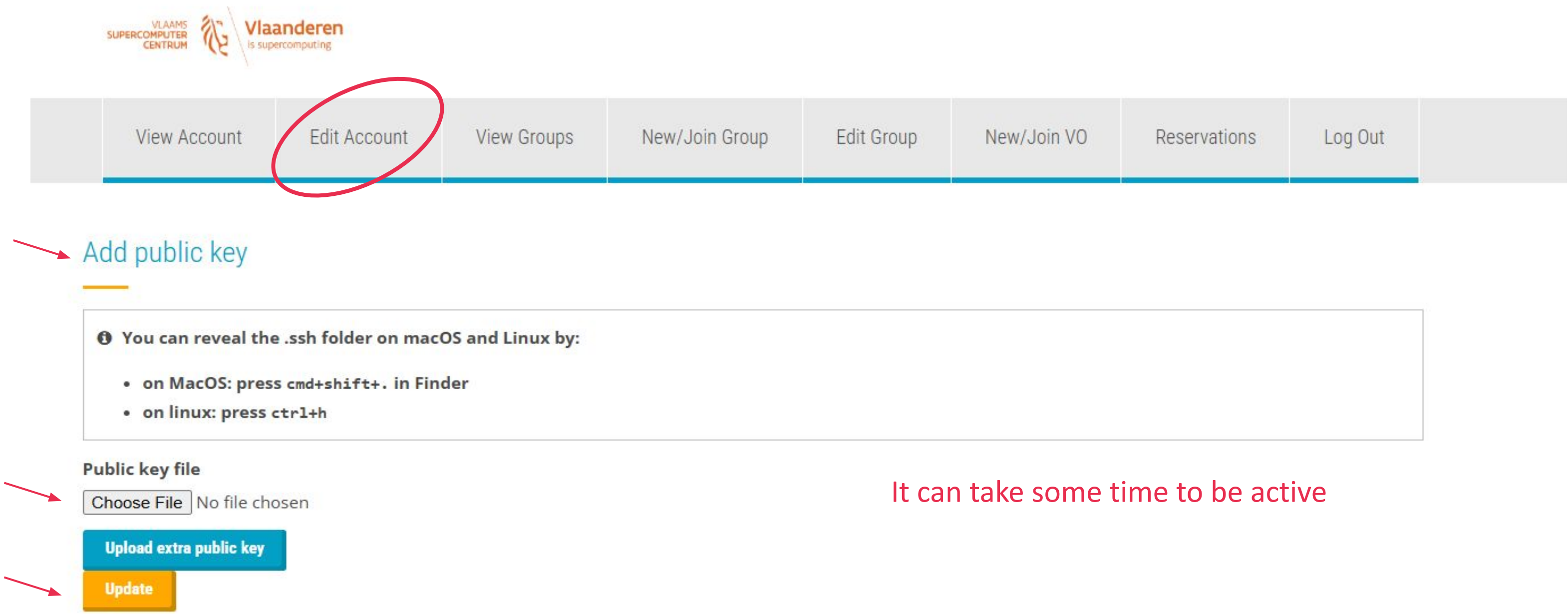
P.S.: Choose a strong password

- Check your new key

```
ls ~/.ssh
```

Inform the key to VSC

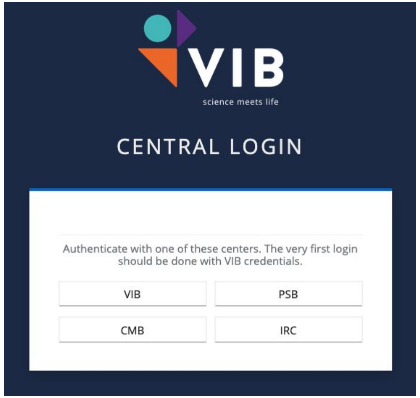
- Access your account
  https://account.vscentrum.be/django//



It can take some time to be active

Authentication for VIB Data Core Compute Cluster

- Guide on **docs.datacore.vib.be**/entrypoints/command-line-access/
- Smallstep CLI leverages VIB SSO for a secure connection



- Generates certificate that is valid for 16 hours

Access from WSL/LINUX terminal

Log in

- VSC

  ssh vsc00000@login.hpc.kuleuven.be

  ssh vsc00000@login.hpc.ugent.be

- VIB

  ssh -p 2022 firstname.lastname@compute.vib.be

We will try later

We need to wait the key to be authorized

Ondemand

Working & checking resources

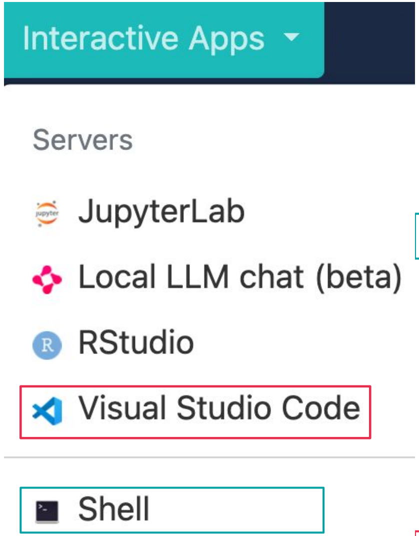
Let’s connect

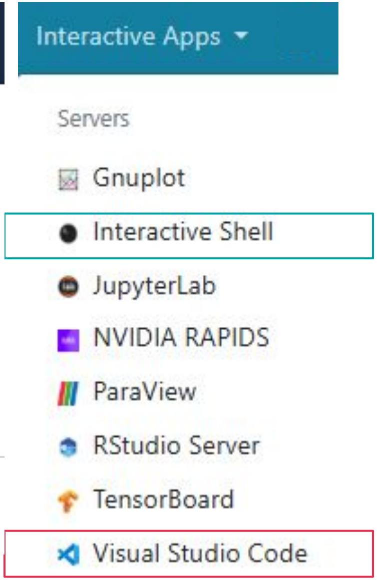
Chose one or both:

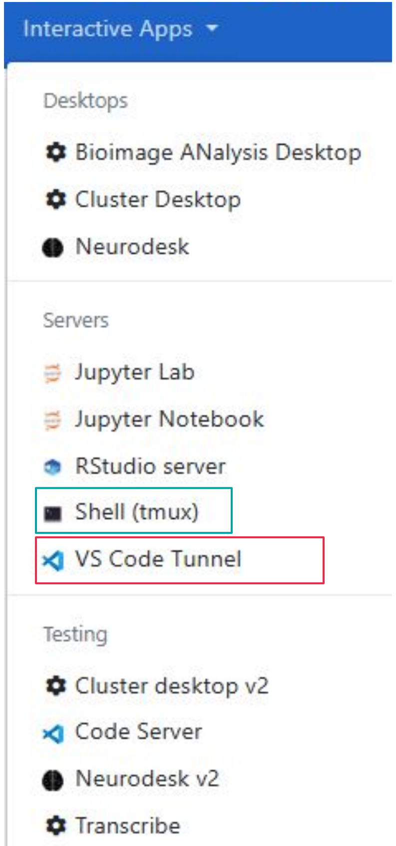
[VIB]()
[UGent Tier-2]()

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP







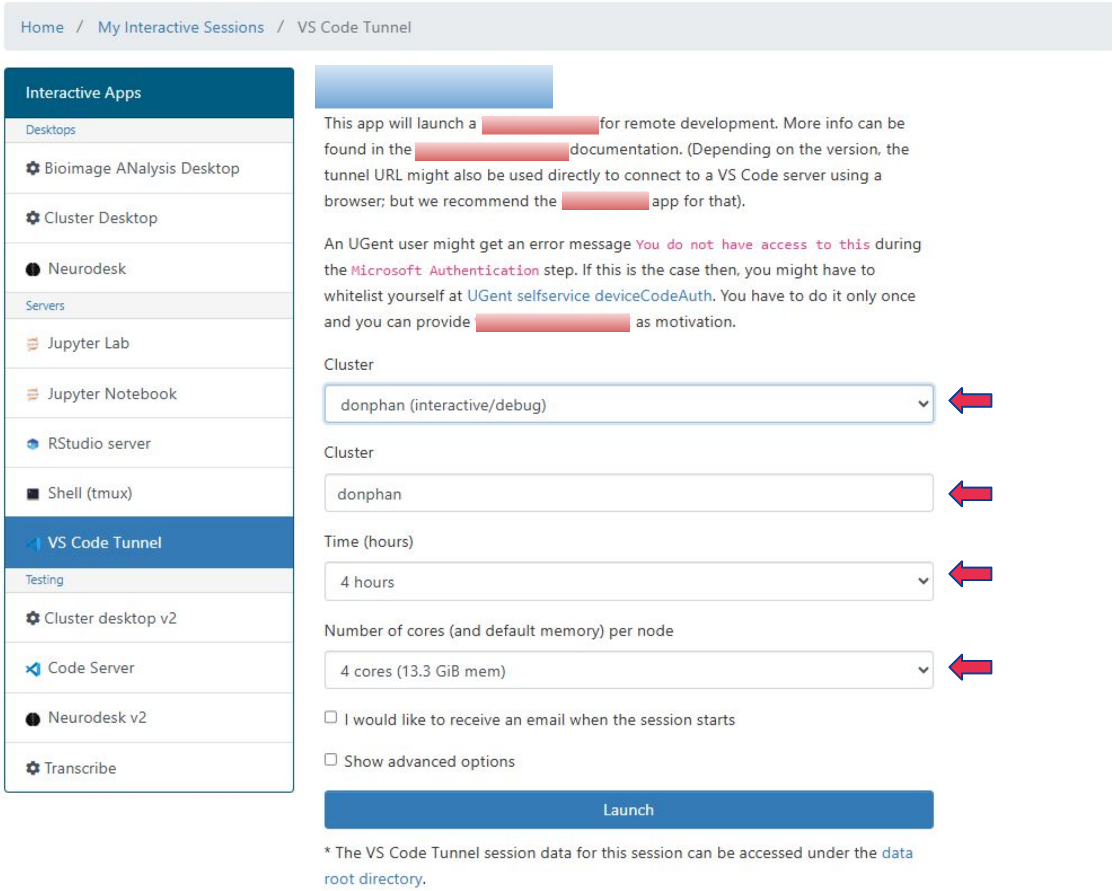
Let’s connect

Chose one or both:

[VIB]
[UGent Tier-2]

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
  - Define resources



Let’s connect

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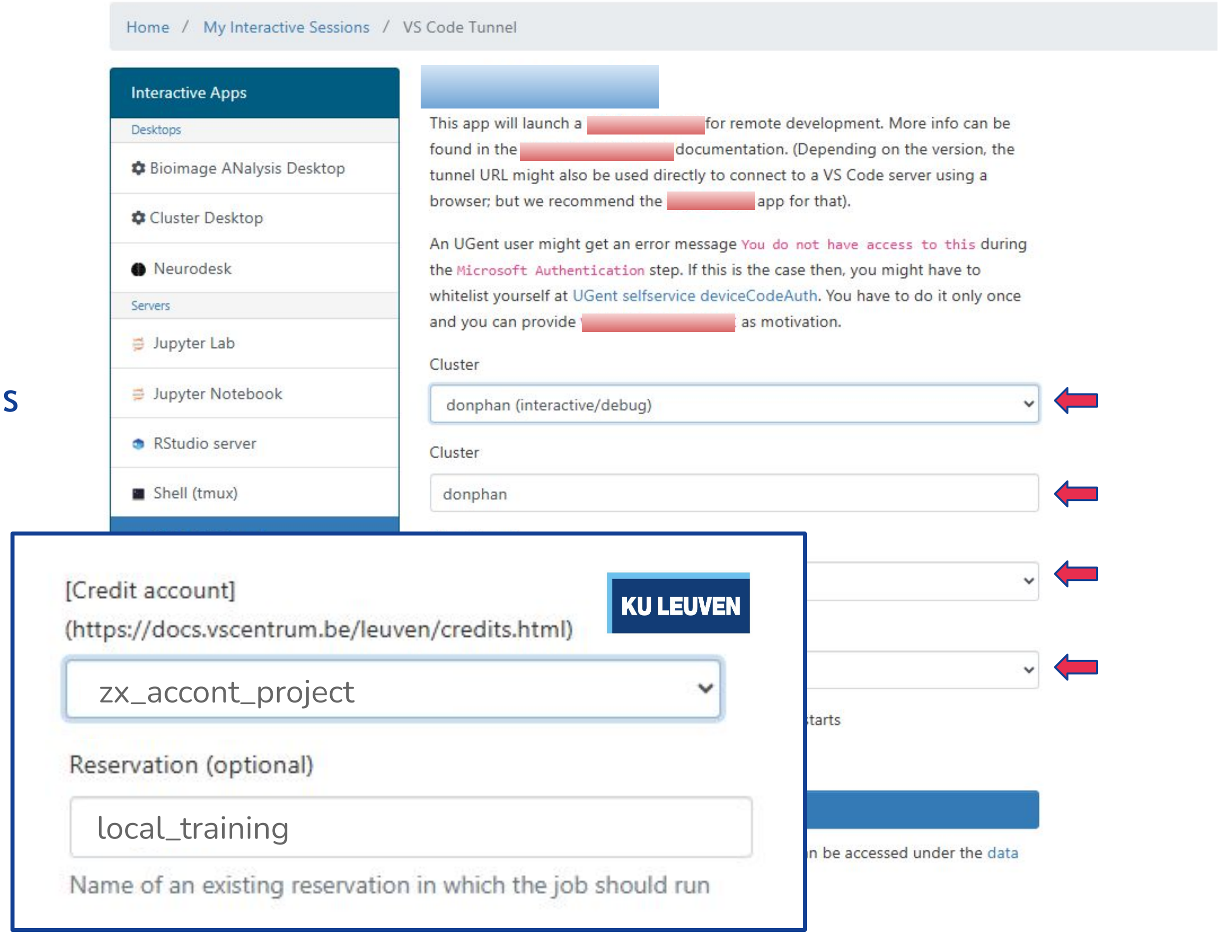


Let’s connect

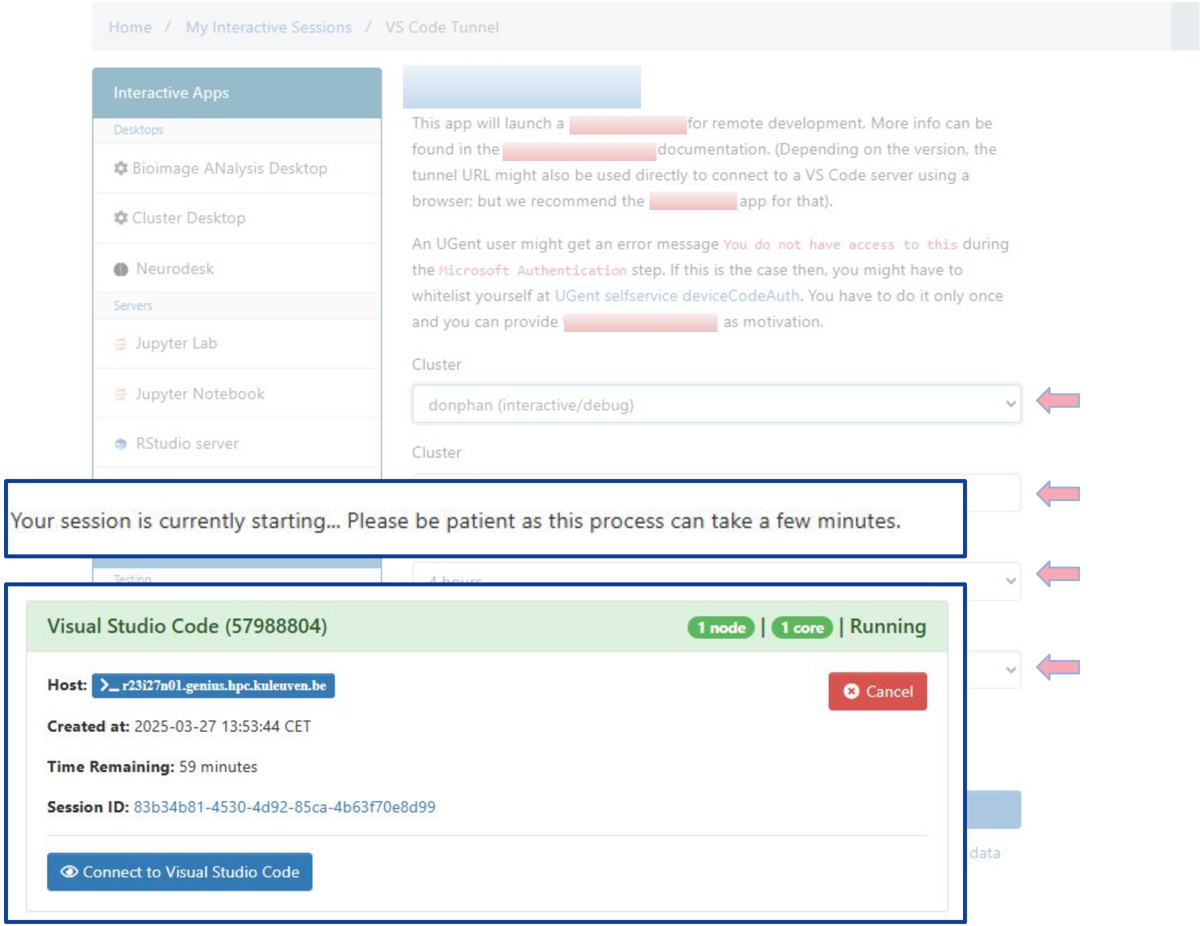
Chose one or both:

[VIB](#)
[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
    - Define resources

Launch → wait …



VIB

Let's connect

login.hpc.ugent.be/node/node4013.donphan.os/13465/?arg=RxxpXAYJGAcfmdZDTbm9

```
[vsc47776@node4013 ~]$ export VSCODE_CLI_USE_FILE_KEYCHAIN=1
[vsc47776@node4013 ~]$ tmux -S /tmp/J20159863.sock wait-for -S CUSTOMCHANNEL
[vsc47776@node4013 ~]$ reset
[vsc47776@node4013 ~]$ echo

[vsc47776@node4013 ~]$ echo "Copy the device code, open the url and paste it"
Copy the device code, open the url and paste it
[vsc47776@node4013 ~]$ echo

[vsc47776@node4013 ~]$ code tunnel --accept-server-license-terms --name ${TUNNEL_NAME:0:20}
*
* Visual Studio Code Server
*
* By using the software, you agree to
* the Visual Studio Code Server License Terms (https://aka.ms/vscode-server-license) and
* the Microsoft Privacy Statement (https://privacy.microsoft.com/en-US/privacystatement).
*
✓ How would you like to log in to Visual Studio Code? · Microsoft Account
```

UNIVERSITEIT GENT

VIB

Let's connect

login.hpc.ugent.be/node/node4013.donphan.os/13465/?arg=RxxpXAYJGAcfmdZDTbm9

```
[vsc47776@node4013 ~]$ export VSCODE_CLI_USE_FILE_KEYCHAIN=1
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[vsc47776@node4013 ~]$ code tunnel --accept-server-license-terms --name ${TUNNEL_NAME:0:20}
*
* Visual Studio Code Server
*
* By using the software, you agree to
* the Visual Studio Code Server License Terms (https://aka.ms/vscode-server-license) and
* the Microsoft Privacy Statement (https://privacy.microsoft.com/en-US/privacystatement).
*
✓ How would you like to log in to Visual Studio Code? · Microsoft Account
To sign in, use a web browser to open the page https://microsoft.com/devicelogin and enter the code LYWC6GL6Y to authenticate.
LYWC6GL6Y[2025-03-27 14:25:13] info Creating tunnel with the name: vsc-vsc47776-donphan
```

UNIVERSITEIT GENT

VIB

Let's connect

login.hpc.ugent.be/node/node4013.donphan.os/13465/?arg=RxxpXAYJGAcfmdZDTbm9

```
[vsc47776@node4013 ~]$ export VSCODE_CLI_USE_FILE_KEYCHAIN=1
[vsc47776@node4013 ~]$ tmux -S /tmp/J20159863.sock wait-for -S CUSTOMCHANNEL
[vsc47776@node4013 ~]$ reset
[vsc47776@node4013 ~]$ echo

[vsc47776@node4013 ~]$ echo "Copy the device code, open the url and paste it"
Copy the device code, open the url and paste it
[vsc47776@node4013 ~]$ echo

[vsc47776@node4013 ~]$ code tunnel --accept-server-license-terms --name ${TUNNEL_NAME:0:20}
*
* Visual Studio Code Server
*
* By using the software, you agree to
* the Visual Studio Code Server License Terms (https://aka.ms/vscode-server-license) and
* the Microsoft Privacy Statement (https://privacy.microsoft.com/en-US/privacystatement).
*
✓ How would you like to log in to Visual Studio Code? · Microsoft Account
To sign in, use a web browser to open the page https://microsoft.com/devicelogin and enter the code LYWC6GL6Y to authenticate.
LYWC6GL6Y[2025-03-27 14:25:13] info Creating tunnel with the name: vsc-vsc47776-donphan

Open this link in your browser https://vscode.dev/tunnel/vsc-vsc47776-donphan/kyukon/home/gent/477/vsc47776

_
```

UNIVERSITEIT GENT

VIB

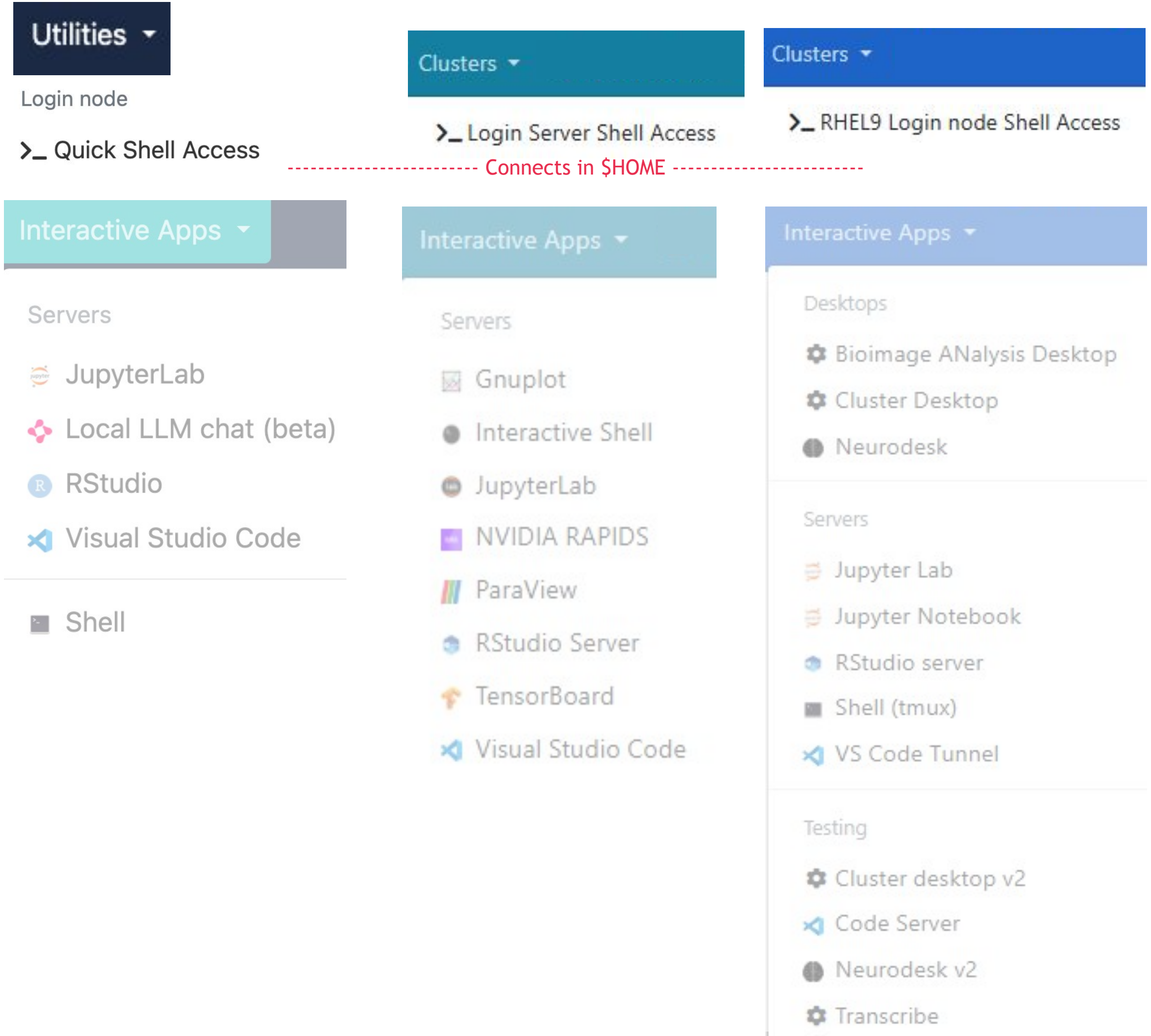
Let’s connect

Chose one or both:

[VIB](#)
[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
  - Define resources
- Choose the Shell
  - Start in $HOME



VIB

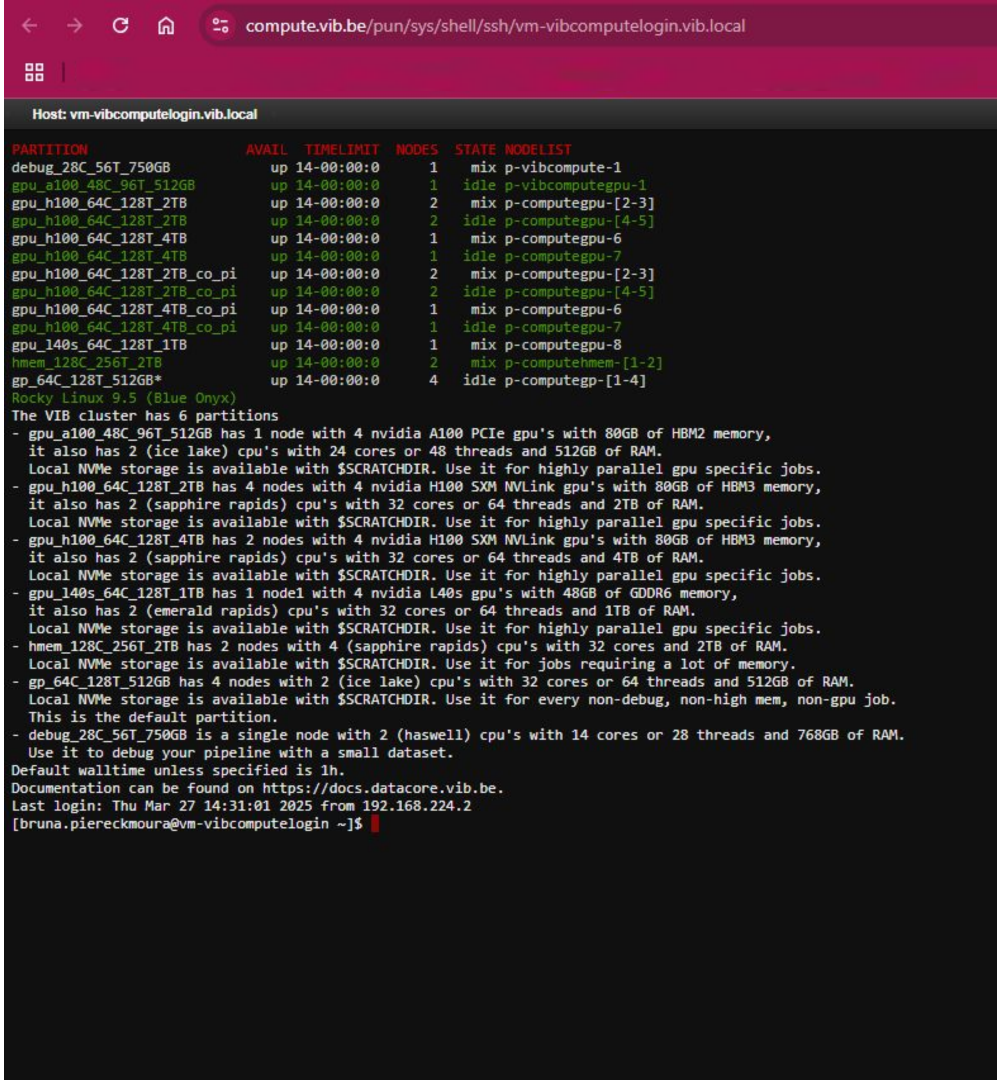
Let’s connect

Chose one or both:

[VIB](#)
[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
  - Define resources
- Choose the Shell
  - Start in $HOME
  - Move to cluster
    - Module swap <cluster>



HPC

Tools available

Terminal Tools

Search for tools:

```
$ module avail <name>
$ module av <name>
$ module spider <name>
```

- D = default
- E = extension

```
[vsc47776@node4016 ~]$ module avail blast

------------------------------------------ /apps/gent/RHEL9/cascadelake-ib/modules/all ------------------------------------------
   BLAST+/2.14.1-gompi-2023a    BLAST+/2.16.0-gompi-2023b    BLAST+/2.16.0-gompi-2024a (D)    crb-blast/0.6.9-GCC-12.3.0

-------------------- This is a list of module extensions. Use "module --nx avail ..." to not show extensions. --------------------
    crb-blast (E)

These extensions cannot be loaded directly, use "module spider extension_name" for more information.

  Where:
   D:  Default Module
   E:  Extension that is provided by another module

If you need software that is not listed, request it via https://www.ugent.be/hpc/en/support/software-installation-request
```

Terminal Tools

Search for tools:

```
$ module avail <name>
$ module av <name>
$ module spider <name>
```

- D = default
- E = extension

Load and use

```
$ module load <name>
```

```
[vsc47776@node4016 ~]$ module load BLAST+/2.16.0-gompi-2024a
[vsc47776@node4016 ~]$ _
```

```
[vsc47776@node4016 ~]$ module avail blast

------------------------------------------- /apps/gent/RHEL9/cascadelake-ib/modules/all -------------------------------------------
   BLAST+/2.14.1-gompi-2023a    BLAST+/2.16.0-gompi-2023b    BLAST+/2.16.0-gompi-2024a (D)    crb-blast/0.6.9-GCC-12.3.0

-------------------- This is a list of module extensions. Use "module --nx avail ..." to not show extensions. --------------------
    crb-blast (E)

These extensions cannot be loaded directly, use "module spider extension_name" for more information.

  Where:
   D:  Default Module
   E:  Extension that is provided by another module

If you need software that is not listed, request it via https://www.ugent.be/hpc/en/support/software-installation-request
```

Terminal Tools

Check loaded modules:

```
$ module list <name>
```

- Modules will be available
  - in the node
  - during time requested

Unload modules

```
$ module unload <name>
$ module purge
```

Terminal Tools

Check loaded modules:

```
$ module list <name>
```

- Modules will be available
  - in the node
  - during time requested

Unload modules

```
$ module unload <name>
$ module purge
```

Some resources are promptly available

If you want to do a request, check if a module exists to facilitate:
https://docs.easybuild.io/api/easybuild/tools/modules/

Terminal Tools

Automatically available:

- Example:
  - python
  - singularity
  - apptainer
  - perls
  - sqlite
  - git

Terminal Tools

VIB Data Core Compute Cluster

- Module lister

Files ▾ Utilities ▾ Interactive Apps ▾ My Interactive Sessions   Develop ▾ Help ▾ Logged in as

Modules Lister

This page lists the **installed software modules** available on the VIB Data Core Compute Cluster via the Lmod module system. Modules are grouped by software tool and list available versions. Information about the modules listed below are provided by EasyBuild, the framweork we use to install modules.

If you'd like a specific module installed, you can contact our support team. Alternatively, you're welcome to install your own local modules following our documentation.

Show 25 entries   Search:

Module	Category	Versions	Description
Abseil	lib	20230125.2-GCCcore-12.2.0 20230125.3-GCCcore-12.3.0 20240116.1-GCCcore-13.2.0 20240722.0-GCCcore-13.3.0	Abseil is an open-source collection of C++ library code designed to augment the C++ standard library. The Abseil library code is collected from Google's own C++ code base, has been extensively tested and used in production, and is the same code we depend on in our daily coding lives.
absl-py	tools	2.1.0-GCCcore-13.3.0	absl-py is a collection of Python library code for building Python applications. The code is collected from Google's own Python code base, and has been extensively tested and used in production.
ACTC	lib	1.1-GCCcore-11.3.0	ACTC converts independent triangles into triangle strips or fans.
adjustText	tools	0.7.3-foss-2021b	A small library for automatically adjustment of text position in matplotlib plots to minimize overlaps.
aiohttp	lib	3.10.10-GCCcore-13.3.0 3.8.1-GCCcore-11.2.0 3.8.3-GCCcore-11.3.0 3.8.5-GCCcore-12.2.0 3.8.5-GCCcore-12.3.0	Asynchronous HTTP client/server framework for asyncio and Python.
almosthere	tools	2.0.2-GCCcore-10.2.0	Progress indicator C library. ATHR is a simple yet powerful progress indicator library that works on Windows, Linux, and macOS. It is non-blocking as the progress update is done via a dedicated, lightweight thread, as to not impair the performance of the calling program.

HPC

Submitting Jobs

Shell approach

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

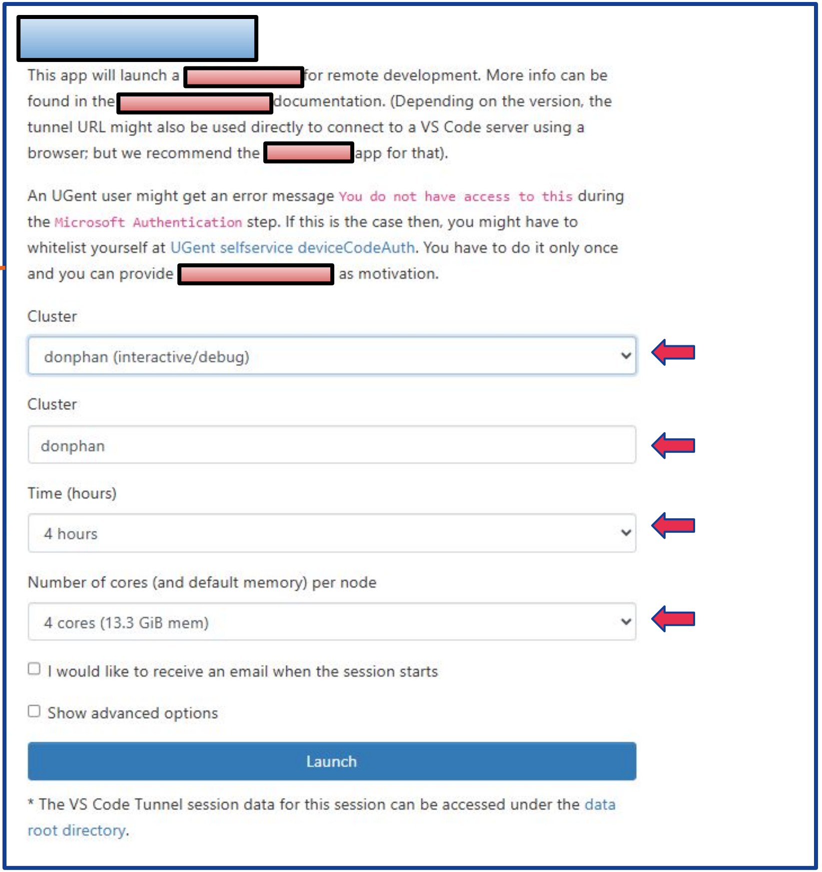
- Arguments
  - Different names
  - Same functionality

Resource managers

- Define resources
  - same-same, not the same !

```
--partition = <cluster_name>
--mem = <number>
--time = hh:mm:ss
--ntasks = <number_cores>
```

  - Scripting = Reproducibility



Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
  - Different names
  - Same functionality

	Slurm	Torque
Submit a job	sbatch script.sh	qsub script.pbs

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
  - Different names
  - Same functionality

	Slurm	Torque
Submit a job		
	sbatch script.sh	qsub script.pbs
Define resources		
	--job-name = <name>	**--N** <job_name>
	--output = <name.out>	**-o** stdout.$PBS_JOBID

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
  - Different names
  - Same functionality

Slurm		Torque
Submit a job		
sbatch script.sh		qsub script.pbs
Define resources		
--job-name = <name>	– – –	--N <job_name>
--output = <name.out>	– – –	**-o** stdout.$PBS_JOBID
--time = hh:mm:ss	– – –	-l walltime = hh:mm:ss
--mem = <n>G	– – –	memory - **-l mem** = <n> [MB] - **-l pmem** = <n> [GB]

VIB

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
  - Different names
  - Same functionality

Slurm	Torque
Submit a job	
sbatch script.sh	qsub script.pbs
UNIVERSITEIT GENT	UNIVERSITEIT GENT
VUB	VUB
KU LEUVEN	
VIB	
University of Antwerp	

Our first job

- Define resources
- Define variables
- Think of modules
- Commands

- Run in Slurm and PBS
  - UGent
  - VIB (if possible)

```
#!/bin/bash
#SBATCH --job-name=test_job_submission
#SBATCH --partition=debug_28C_56T_750GB
#SBATCH --mem=8G
#SBATCH --time=00:30:00

# Define variables
DOCKER_IMG="docker://hello-world"

# Practice best practice by purging all loaded modules first
module purge

# Load necessary modules (none needed for this example)
# module load Python/3.13.1-GCCcore-14.2.0

# Pull minimal Docker image using Singularity
singularity pull $DOCKER_IMG

# Sleep for 20 seconds so there is time to see the job via `squeue`
echo 'Going to sleep for 20 seconds... ZZZzzzzz....'
sleep 20
```

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Common tasks:

- Cancel a job
  - Scancel <id>
  - qdel <id>
- List Jobs
  - qstat
  - squeue

Slurm	Torque
Submit a job	
sbatch script.sh	qsub script.pbs
Cancel a job	
scancel <id>	qdel
List jobs	
squeue	qstat

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Common tasks:

- Cancel a job
  - Scancel <id>
  - qdel <id>
- List Jobs
  - qstat
  - squeue

seff <jobID>
Slurm_jobinfo <jobID>

	Slurm	Torque
Submit a job	sbatch script.sh	qsub script.pbs
Cancel a job	scancel <id>	qdel
List jobs	squeue	qstat
	Status possible: *R: running* *CD: completed* S: suspended PD: pending CG: completing	Status possible: *R: running* *C: completed* S: suspended Q: queued E: exiting

VIB

HPC

Connect cmd-line

Access from WSL/LINUX terminal

- Log in

  - ssh vsc00000@login.hpc.kuleuven.be
  - ssh vsc47776@login.hpc.ugent.be
  - ssh -p 2022 firstname.lastname@compute.vib.be

It can take some time to be active

Bruna Piereck

bruna.piereckmoura@vib.be

Primary color palette

Aqua
HEX #02A2A7
RBG 2, 162, 167

Orange
HEX #EC6523
RBG 236, 101, 35

Purple
HEX #693A91
RBG 105, 58, 145

Navy
HEX #14274C
RBG 20, 39, 76

Tint 01
HEX #29B3B8
RBG 41, 179, 184

Tint 01
HEX #EF7739
RBG 239, 119, 57

Tint 01
HEX #784B9A
RBG 120, 75, 154

Tint 02
HEX #99D7D6
RBG 153, 215, 214

Tint 02
HEX #F7C1A3
RBG 247, 193, 163

Tint 02
HEX #C2AFD0
RBG 194, 175, 208

Tint 03
HEX #CBEAEC
RBG 203, 234, 236

Tint 03
HEX #FBE8DF
RBG 251, 232, 223

Tint 03
HEX #E3DCEA
RBG 227, 220, 234

Secondary color palette

Blue
HEX #1F448F
RBG 31, 68, 143

Cherry
HEX #E83050
RBG 232, 48, 80

Green
HEX #00A14A
RBG 0, 161, 74

Yellow
HEX #F9B418
RBG 249, 180, 24

Tint 01
HEX #24539F
RBG 36, 83, 159

Tint 01
HEX #EC4D6A
RBG 236, 77, 106

Tint 01
HEX #1AAF5E
RBG 26, 175, 94

Tint 01
HEX #FCC747
RBG 252, 199, 71

Tint 02
HEX #A1B2D5
RBG 161, 178, 213

Tint 02
HEX #F6AAB9
RBG 246, 170, 185

Tint 02
HEX #9BD3B3
RBG 155, 211, 179

Tint 02
HEX #FCE39B
RBG 252, 227, 155

Tint 03
HEX #D5DBEC
RBG 213, 219, 236

Tint 03
HEX #FCE3E6
RBG 252, 227, 230

Tint 03
HEX #D1EAD9
RBG 209, 234, 217

Tint 03
HEX #FEF4CF
RBG 252, 244, 207

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